

House Price Prediction – Summary Report

Objective

The goal of this project was to develop machine learning models capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and location-related features. The project also aimed to identify the factors that most strongly influence housing prices.

Data Preparation

The Housing dataset was loaded and explored using Pandas. Missing values and duplicate records were checked and handled appropriately. Categorical variables such as main road access, guest room availability, air conditioning, and furnishing status were converted into numerical form using one-hot encoding to make them suitable for machine learning algorithms.

Models Used

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was split into 80% training data and 20% testing data. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score.

Results

Metric	Linear Regression	Random Forest
MAE	214,070	283,662
RMSE	269,486	364,224
R ² Score	0.9554	0.9186

The Linear Regression model outperformed the Random Forest model on this dataset, achieving an R² score of 95.54%, which indicates that the model explains more than 95% of the variation in house prices.

Key Findings

Feature importance analysis revealed that property area is the most influential factor affecting house prices. The area feature accounted for approximately 89% of the overall predictive importance in the Random Forest model. Preferred-area classification also showed a strong positive relationship with property value.

Interestingly, amenities such as hot water heating, guest rooms, and furnishing status had a much smaller impact on price than expected. The results suggest that physical property size and location contribute far more to housing value than additional lifestyle features.

Business Recommendation

Real estate companies can improve pricing accuracy by implementing automated valuation systems that primarily focus on area and preferred-area indicators. These features alone provide highly reliable price estimates and can significantly reduce manual valuation effort while supporting faster decision-making for buyers and sellers.

Conclusion

The project successfully developed an accurate house price prediction system. Among the models tested, Linear Regression provided the best performance due to the largely linear relationships present in the dataset. The findings demonstrate that property size and location remain the dominant drivers of residential real estate prices.